Huawei Access Node ELEC-ACC-57 Routing Protocol Flap Resolution SOP

Node: ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3538

Description

In a Huawei Access Node, such as ELEC-ACC-57 at the Electronic City Hub, routing protocol flap failures (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) are precipitated by transient instability in the control plane, primarily stemming from misconfigurations, hardware malfunctions, or transient network conditions. This instability causes the routing protocols to rapidly oscillate between states of convergence and divergence, resulting in a cascade of routing table updates, advertisements, and withdrawals that can lead to network instability and packet loss. The alarm code ALARM_3538 is triggered when the system detects a critical instance of routing protocol flap, indicating a high risk of network disruption. As the access node is a critical component of the Electronic City Hub, prompt resolution of this issue is imperative to prevent network downtime and ensure continuous service availability.

Pre-requisite Checks

- Verify that the Electronic City Hub is accessible and that the network environment is stable.

- Check the status of the redundant links and ensure they are functioning correctly.

- Obtain clearance from the Network Operations Center (NOC) to proceed with troubleshooting and resolution activities.

Potential Root Causes

Cause	Probability
Misconfiguration of routing protocols	High
Hardware malfunction in the access node	Medium
Transient network conditions	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: BGP process information, Description: This command displays BGP process information, including the BGP version, AS number, and peer group configuration. It helps identify any misconfigurations or anomalies in the BGP configuration.

Command: display ospf peer, Expected Output: OSPF peer information, Description: This command displays OSPF peer information, including the OSPF version, process ID, and neighbor adjacency status. It helps identify any issues with OSPF adjacency or neighbor relationships.
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```
Command: display isis lsp, Expected Output: ISIS LSP information, Description: This command displays ISIS LSP information, including the ISIS version, process ID, and LSP flooding status. It helps identify any issues with ISIS LSP flooding or misconfigurations.
```

Resolution Steps

Step 1: Step: Verify BGP configuration, Command: display bgp configuration, Description: Verify the BGP configuration, including the AS number, peer group configuration, and neighbor IP addresses. Ensure that the BGP configuration is correct and consistent with the network design.

Step 2: Step: Disable BGP, Command: undo bgp all, Description: Disable the BGP process on the access node to prevent any further routing protocol flap. This step is critical to prevent network instability and packet loss.

Step 3: Step: Check OSPF configuration, Command: display ospf configuration, Description: Verify the OSPF configuration, including the OSPF version, process ID, and neighbor IP addresses. Ensure that the OSPF configuration is correct and consistent with the network design.

Step 4: Step: Restart OSPF, Command: restart ospf, Description: Restart the OSPF process on the access node to ensure that the OSPF adjacency is re-established correctly.

Step 5: Step: Verify ISIS configuration, Command: display isis configuration, Description: Verify the ISIS configuration, including the ISIS version, process ID, and LSP flooding status. Ensure that the ISIS configuration is correct and consistent with the network design.

Post-Check Verification

```
Command: display bgp process, Expected Output: BGP process information, Description: Verify that the BGP process is stable and converging correctly.

Command: display ospf peer, Expected Output: OSPF peer information, Description: Verify that the OSPF adjacency is established correctly and that the OSPF process is stable.

Command: display isis lsp, Expected Output: ISIS LSP information, Description: Verify that the ISIS LSP flooding is stable and that the ISIS process is converging correctly.
```

Roll-back Procedures

- Step: Re-enable BGP, Command: bgp all, Description: Re-enable the BGP process on the access node to restore routing protocol functionality.

- Step: Re-start OSPF, Command: restart ospf, Description: Re-start the OSPF process on the access node to restore OSPF adjacency and stability.